

1200EET304052301

| APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY                                  |    |                                                                                                                                                                 |       |
|---------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| SIXTH SEMESTER B.TECH DEGREE(R,S)EXAMINATION, May/July 2023 (2019 Scheme) |    |                                                                                                                                                                 |       |
| Course Code: EET304                                                       |    |                                                                                                                                                                 |       |
| Course Name: POWER SYSTEMS                                                |    | June 2023                                                                                                                                                       |       |
| Max. Marks: 100                                                           |    | Duration: 3 Hours                                                                                                                                               |       |
| PART A                                                                    |    |                                                                                                                                                                 |       |
|                                                                           |    | Answer all questions, each carries 3 marks.                                                                                                                     | Marks |
| 1                                                                         |    | XG1=0.686 pu, XG2=0.457 pu, Xm1=1.025 pu, Xm2=0.683 pu----(2 marks)<br>Diagram-(1 mark)                                                                         | (3)   |
| 2                                                                         |    | Effect of faults in a power system-(1.5 marks), Symmetrical fault-(1.5 marks)                                                                                   | (3)   |
| 3                                                                         |    | Classification of buses – (1.5 Marks)<br>Specifying known and unknown quantities –( 1.5 Marks)                                                                  | (3)   |
| 4                                                                         |    | No. of Non-zero elements in Y-Bus = 10000 x 0.1 =1000 (1 Mark)<br>No. of Non-zero mutual admittances = 900 (1 Mark)<br>No. of Transmission Lines = 450 (1 Mark) | (3)   |
| 5                                                                         |    | $\delta_0=\sin^{-1}(0.75)=48.59$ (1 mark) $(\partial P_e/\partial \delta)\delta_0=1.322$ -(1 mark)<br>$f_n=1.326$ Hz-(1 mark)                                   | (3)   |
| 6                                                                         |    | Definition of synchronizing coefficient- (1mark), Equation- (1mark) Value of synchronizing coefficient the system remains stable.- (1mark)                      | (3)   |
| 7                                                                         |    | Any 3 points-(3 marks)                                                                                                                                          | (3)   |
| 8                                                                         |    | Differences-(3 marks)                                                                                                                                           | (3)   |
| 9                                                                         |    | Explanation of thermal unit constraint-(3 marks)                                                                                                                | (3)   |
| 10                                                                        |    | Definition: – (1.5 Marks), Equation – (1.5 Marks)                                                                                                               | (3)   |
| PART B                                                                    |    |                                                                                                                                                                 |       |
| Answer one full question from each module, each carries 14 marks.         |    |                                                                                                                                                                 |       |
|                                                                           |    | Module I                                                                                                                                                        |       |
| 11                                                                        | a) | Derivation of the expression for fault current in single line to ground fault on unloaded generator. : (7 marks)<br>Drawing : (3 marks.)                        | (10)  |
|                                                                           | b) | $I_c=7.65\angle -157.4^{\circ}$ (1 mark)<br>$I_{a0}=0$ (1mark)                                                                                                  | (4)   |

**1200EET304052301**

|    |    |                                                                                                                                                                                                                                                                                                                        |      |
|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|    |    | $I_{a1} = -0.196 - j1.49$ (1 mark)<br>$I_{a2} = 7.26 - j5.57$ (1 mark)<br>Only equations- (1.5 marks)                                                                                                                                                                                                                  |      |
|    |    | <b>OR</b>                                                                                                                                                                                                                                                                                                              |      |
| 12 | a) | Prefault voltage $= 0.9636 \angle 0^\circ$ (1 mark)<br>Prefault current $= 0.7783 \angle 36.9^\circ$ (1 mark)<br>$I_{g''} = 0.6226 - j1.67$ (3 marks)<br>$I_{m''} = -0.6226 - j6.89$ (3 marks)<br>$I_f = -j11.29$ (1 mark)<br><i>Marks may be awarded if problem is solved by any other method as well.</i>            | (10) |
|    | b) | Circuit breaker ratings with expressions and explanation: (4 marks)<br><i>Marks may be awarded if question is attempted.</i>                                                                                                                                                                                           | (4)  |
|    |    | <b>Module II</b>                                                                                                                                                                                                                                                                                                       |      |
| 13 | a) | $Y_{bus}$ (3 marks)<br>Converting to pu: (1 mark)<br>$V_2^1 = 0.97 - j0.04$ (2 marks)<br>$Q = 1.16$ (2 marks)<br>$V_3^1 = 1.05 - j0.07$ (2 marks)                                                                                                                                                                      | (10) |
|    | b) | What is the Jacobian Matrix- (2 marks)<br>How are the elements of the Jacobian computed- (2 marks)                                                                                                                                                                                                                     | (4)  |
|    |    | <b>OR</b>                                                                                                                                                                                                                                                                                                              |      |
| 14 | a) | Determination of $Y_{bus}$ : (3 marks)<br>Assumption of initial values: (1 mark)<br>$V_2^1 = 1.011 - j0.02$ (1 mark)<br>$V_{2acc}^1 = 1.018 - j0.04$ (1 mark)<br>$V_3^1 = 0.994 - j0.02$ (1 mark)<br>$V_{3acc}^1 = 0.99 - j0.046$ (1 mark)<br>$V_4^1 = 0.97 - j0.064$ (1 mark)<br>$V_{4acc}^1 = 0.95 - j0.01$ (1 mark) | (10) |
|    | b) | Any 4 points- (4 marks)                                                                                                                                                                                                                                                                                                | (4)  |
|    |    | <b>Module III</b>                                                                                                                                                                                                                                                                                                      |      |
| 15 | a) | State and explain equal area criterion: (4 marks)<br>Application of equal area criterion to find the maximum additional load that can                                                                                                                                                                                  | (8)  |

**1200EET304052301**

|    |    |                                                                                                                                                                                                                                           |      |
|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|    |    | be suddenly added-(4 marks)                                                                                                                                                                                                               |      |
|    | b) | Stored energy=1200MJ - (2 marks)<br>M=0.1333MJsec/elec deg - (2 marks)<br>Rotor acceleration=300 elec deg/sec <sup>2</sup> - (2 marks)                                                                                                    | (6)  |
|    |    | <b>OR</b>                                                                                                                                                                                                                                 |      |
| 16 | a) | $\alpha=17.458^\circ$ - (1 mark)<br>$V_t=1<17.450$ -(1 mark)<br>$I=1.012<8.70$ -(1 mark)<br>$E'=1.05<28.440$ -(1 marks)<br>a) $P_e=2.10 \sin\delta$ -(1.5 marks)<br>b) $P_e=0.808\sin\delta$ -(1.5 marks)<br>Power angle curves-(1 marks) | (8)  |
|    | b) | Critical clearing angle and time with expressions-(3 marks each)                                                                                                                                                                          | (6)  |
|    |    | <b>Module IV</b>                                                                                                                                                                                                                          |      |
| 17 | a) | Derivation:(7 marks)<br>Block diagram: (3 marks)                                                                                                                                                                                          | (10) |
|    | b) | Energy stored at no load=750MJ-(1 mark)<br>Excessive KE put into rotating parts of machine-25MJ-(1 mark)<br>Equation-(1 mark)<br>$F_{new}=50.83\text{Hz}$ -(1 mark)                                                                       | (4)  |
|    |    | <b>OR</b>                                                                                                                                                                                                                                 |      |
| 18 | a) | Block diagram-(4 marks)<br>Equation-(4 marks)<br>Explanation-(2 marks)                                                                                                                                                                    | (10) |
|    | b) | $R=0.004\text{Hz/MW}$ -(1 mark)<br>Equation $\Delta P = \frac{-\Delta f}{R}$ -(1 mark)<br>Turbine power increase, $\Delta P = 30 \text{ MW}$ -(2 marks)                                                                                   | (4)  |
|    |    | <b>Module V</b>                                                                                                                                                                                                                           |      |
| 19 | a) | Economic Loading $P_1=100.7 \text{ MW}$ , $P_2=124.21\text{MW}$ , -(4 marks)<br>Equal loading $P_1=112.5 \text{ MW}$ and $P_2=112.5 \text{ MW}$ (4 marks)<br>Saving=Rs 11/hr-(2 marks)                                                    | (10) |
|    | b) | $\beta$ -coefficients- (2 marks)                                                                                                                                                                                                          | (4)  |

1200EET304052301

|      |    |                                                                                                                                                                                                                                                                                                                                                                                       |      |
|------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|      |    | Expression-(2 marks)                                                                                                                                                                                                                                                                                                                                                                  |      |
|      |    | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                             |      |
| 20   | a) | $B_{11} = 0.00114 \text{ MW-1}$ - (2 marks)<br>$P_1 = 89.6 \text{ MW}$ - (2 marks)<br>$P_2 = 66.67 \text{ MW}$ - (2 marks)<br>$PL = 9.15 \text{ MW}$ - (2 marks)<br>$PD = 147.12 \text{ MW}$ - (2 marks)<br><i>As Cost function is given instead of incremental cost, if the particular student writes the general equation and method of solving correctly, marks may be awarded</i> | (10) |
|      | b) | Spinning reserve-(2 marks) Reasons-(2 marks)                                                                                                                                                                                                                                                                                                                                          | (4)  |
| **** |    |                                                                                                                                                                                                                                                                                                                                                                                       |      |